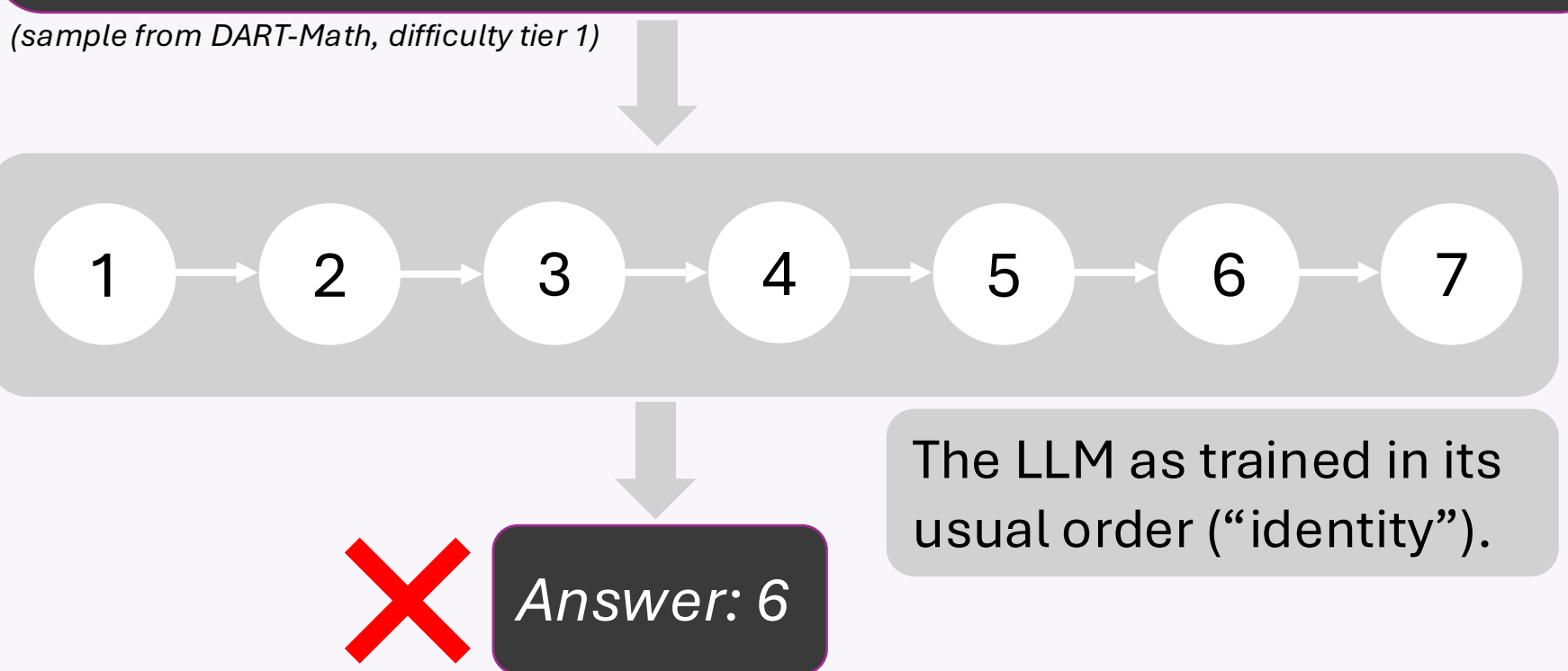


Can we improve transformers by skipping and duplicating their layers?

1. Pre-Study

Can transformer layers cooperate out of their usual order?

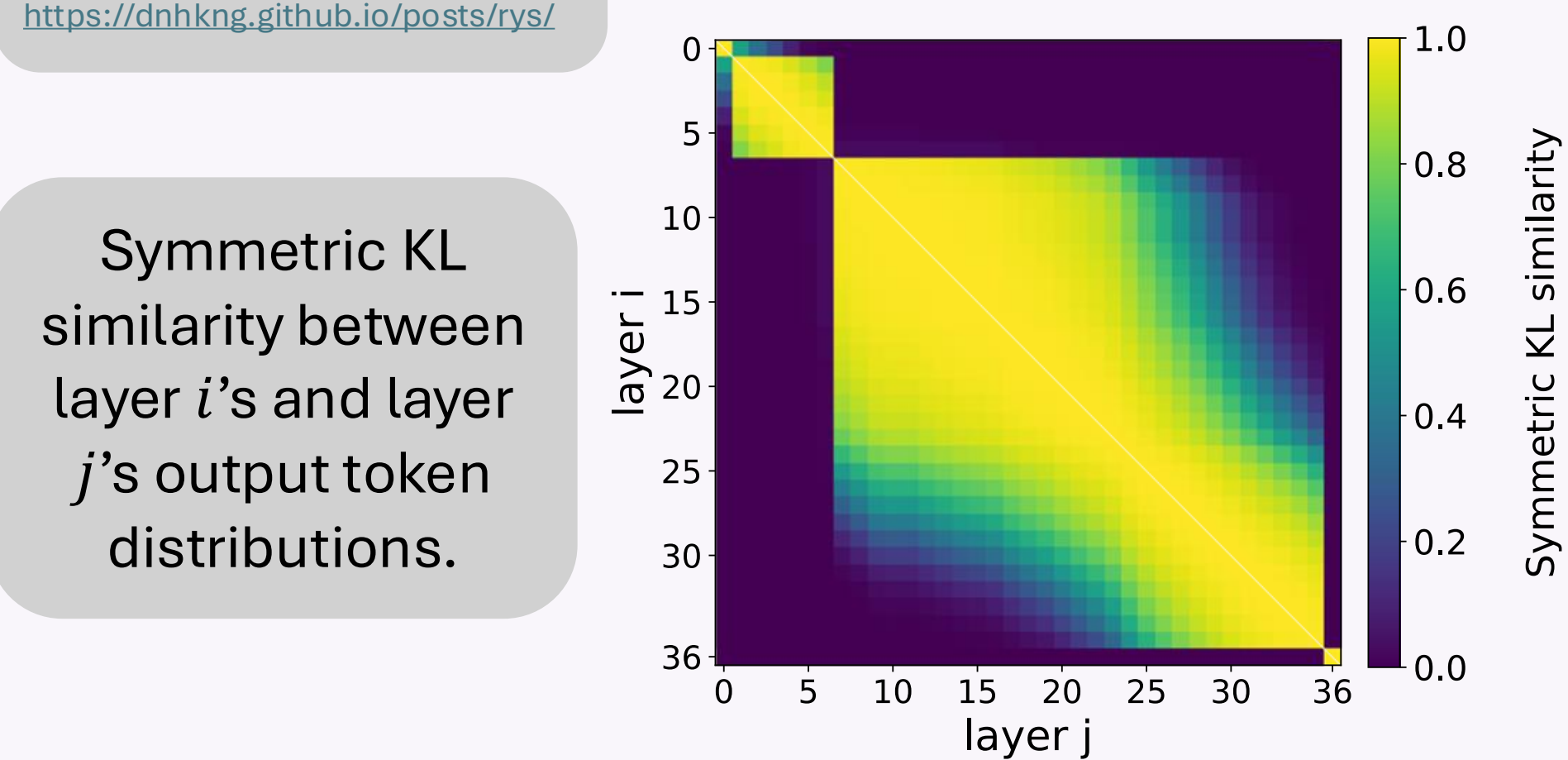
Question: Point A has coordinates (x,6). When Point A is reflected over the y-axis it lands on Point B. What is the sum of the four coordinate values of points A and B?
(sample from DART-Math, difficulty tier 1)



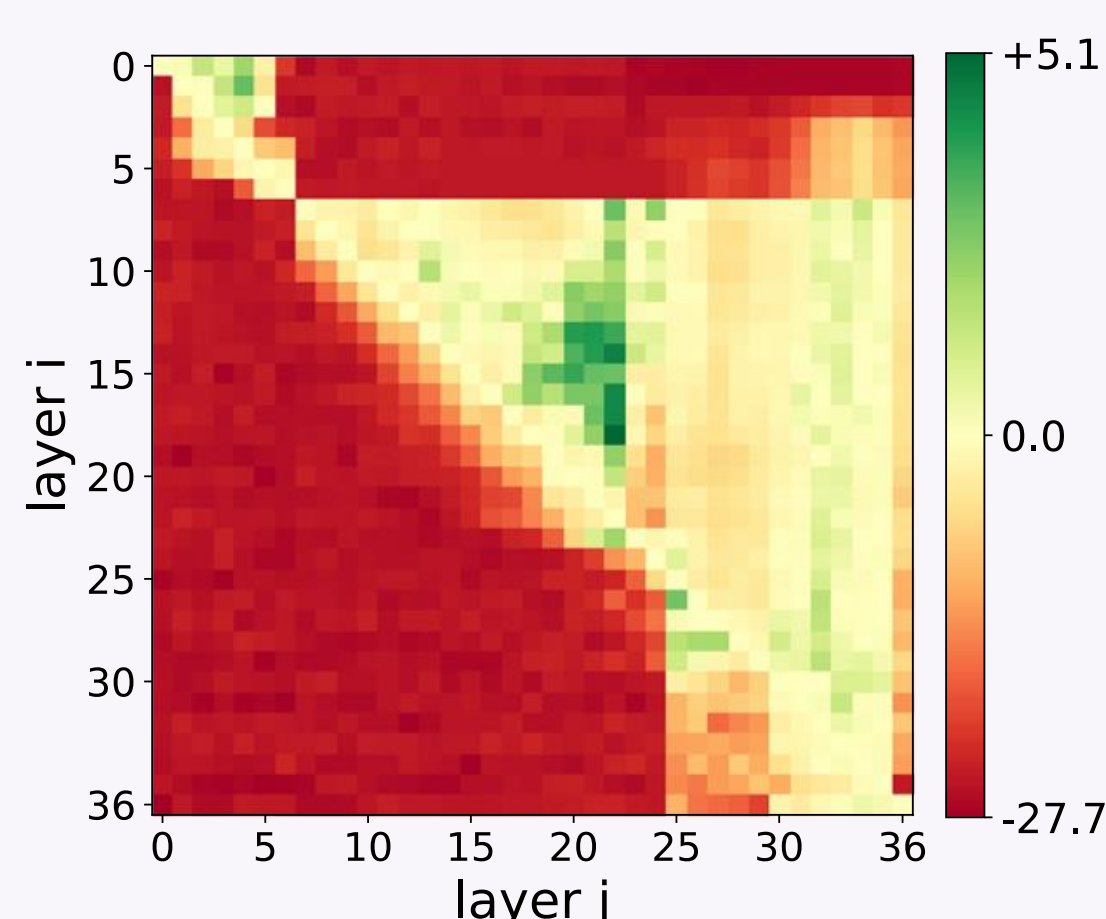
Inspiration: David Noel Ng, *LLM Neuroanatomy: How I Topped the LLM Leaderboard Without Changing a Single Weight*, March 2026, <https://dnhkg.github.io/posts/rys/>

Qwen3-8B improves on a 6-domain subset of MMLU-Pro when repeating a middle block (layers 18–22).

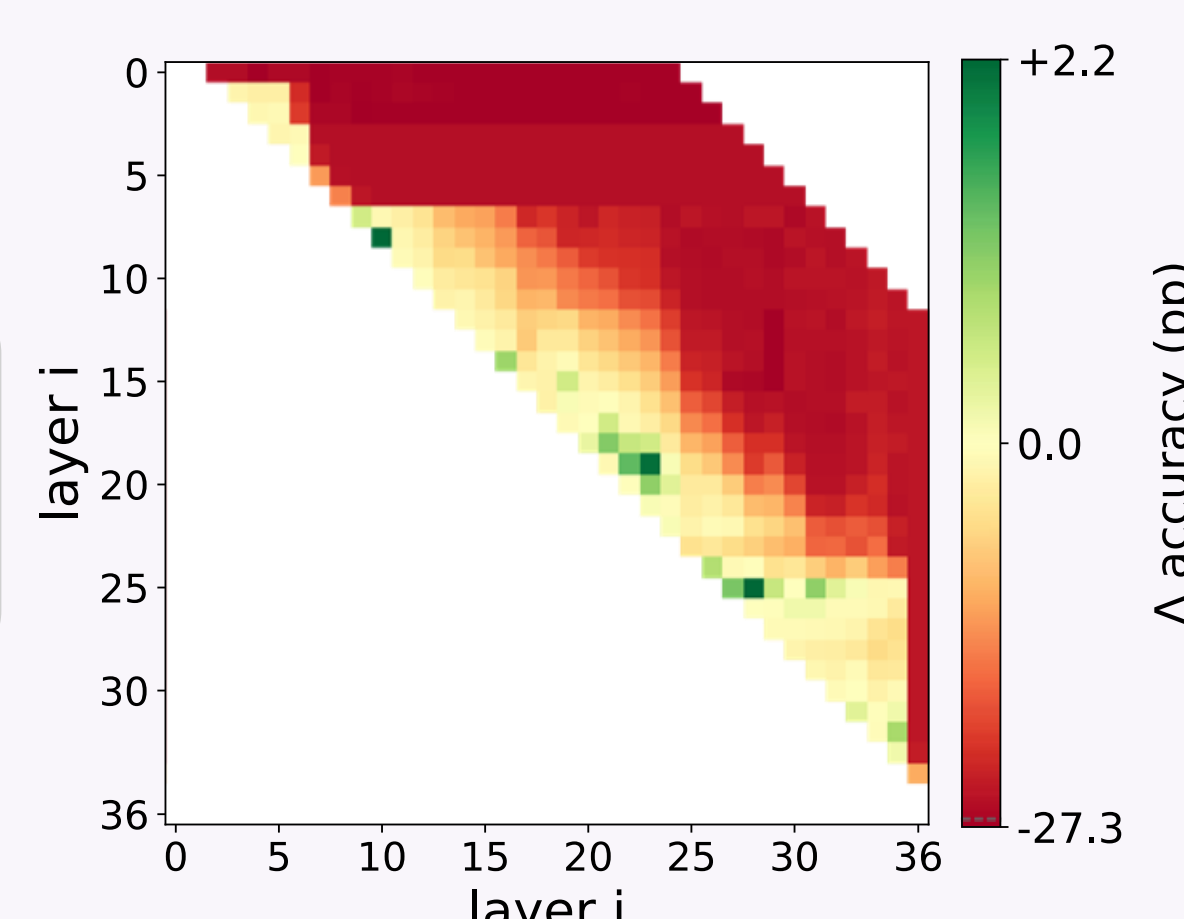
The **representational** and **behavioral** patterns in all three figures agree on two boundaries: layer 6 and layer 25.



Re-running block $[i, j]$ a second time (repeat, upper triangle) or removing it (skip, lower triangle).



Running block $[i, j]$ for $j - i$ times, in parallel.

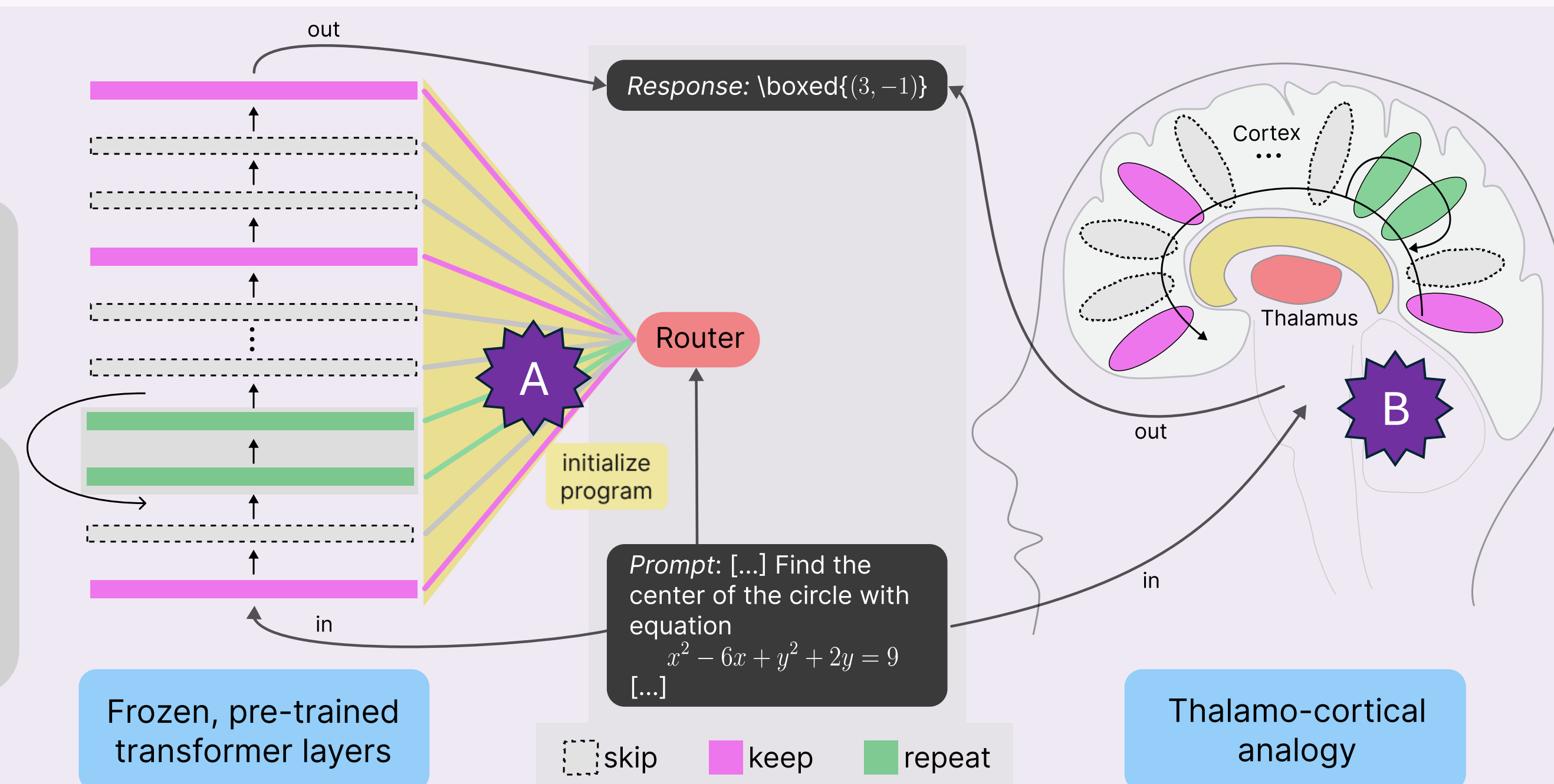


2. Cortical Analogy

Cortical areas as transformer layers.

A Li et al. [1] introduce program-of-layers (**PoLar**), using layers of transformers as a **library of functions** rather than a fixed sequence. They further train a lightweight **router**, that predicts programs for unknown sequences.

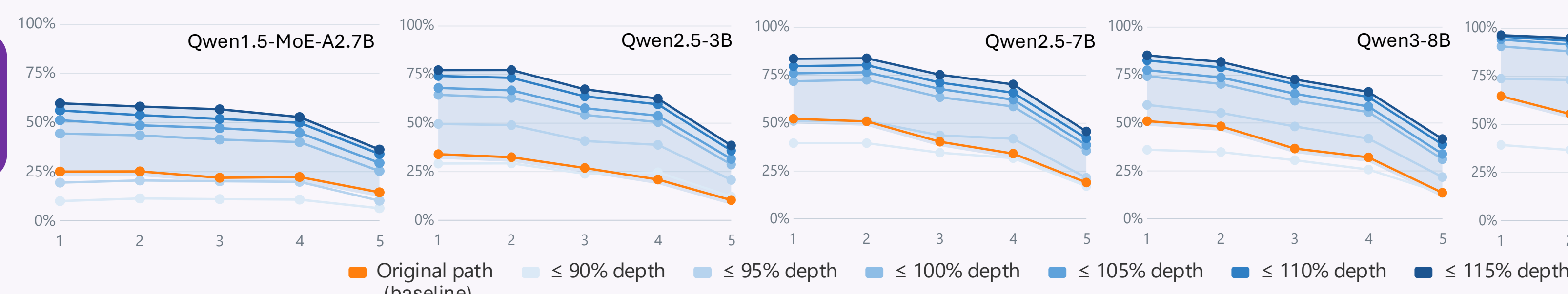
B **Functionally**, we see an analogy to the thalamus routing computation across cortical areas. That cortical areas realize attention-like computation is described in Babu et al. [2]. Larkum [3] argues that a column's architecture and its interaction with the thalamus let the brain integrate context into local computation and distribute the context-sensitive output around the cortex.



3. Programs-of-Layers

Searching programs: Running a Monte-Carlo Tree Search for every sample (here, focused on DART-Math, split into 5 difficulties).

I Baseline is beatable...



II ... even with shorter programs (Occam's Razor)!

Mean and median of the shortest program's executed length (Qwen3-32B).

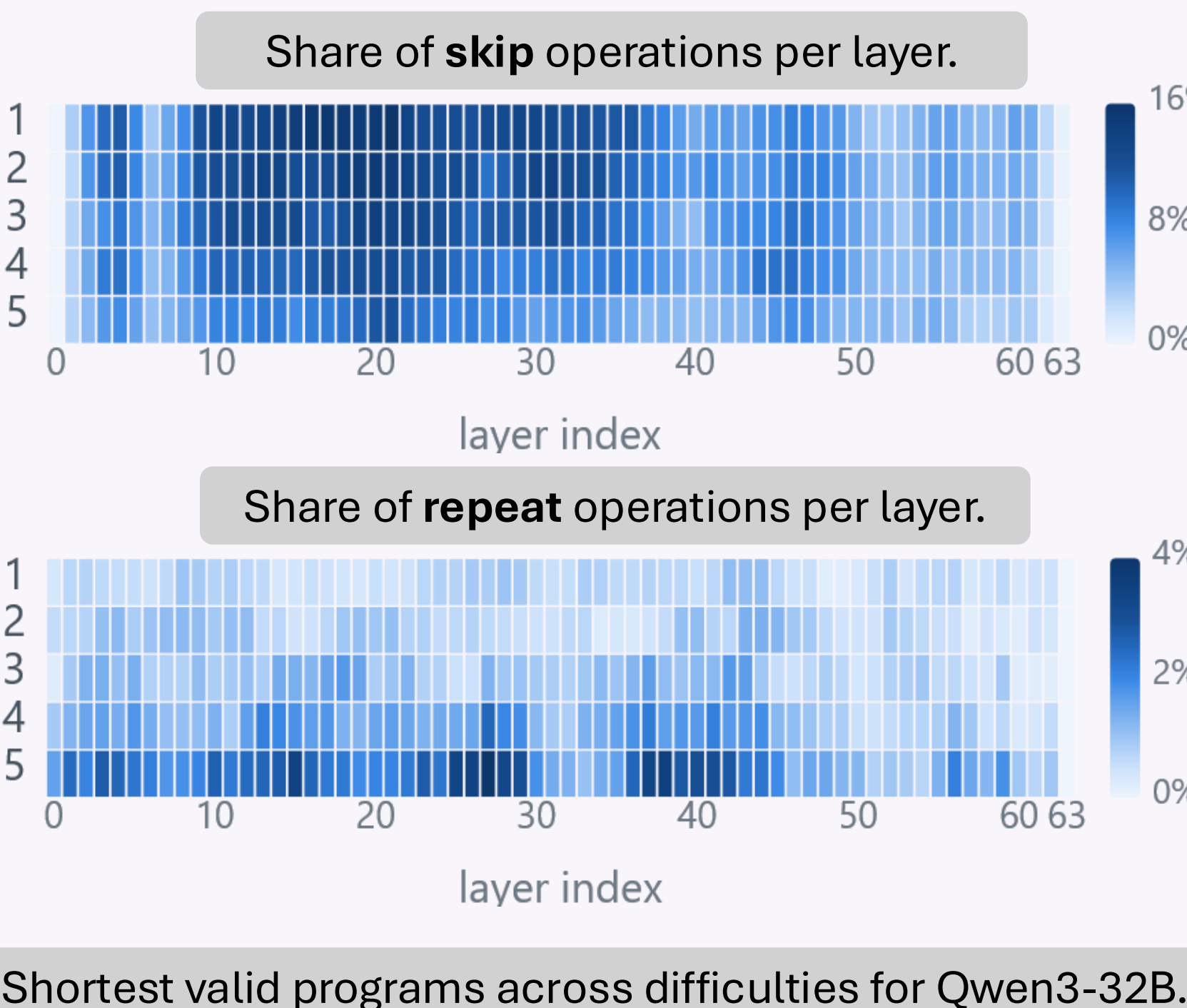


III A few programs solve a lot!

Coverage of the top- k programs (dashed = MCTS baseline).



IV More repetitions for harder programs!



Take-home message
There is a functional similarity between transformer layers and cortical columns.

There's more!

In our paper, we give more detail about the **MCTS**, go deeper into **program robustness**, training a **router** on top of found programs, looping multiple times, and check the method on another dataset.

Very related

Major related works are LayerDrop, LaCo, Mixture-of-Depths, ShortGPT, LayerSkip, FlexiDepth, MindSkip, GateSkip, **DR.LLM**, and, of course, **PoLar** [1] itself.